

PUMPS

- Pump is a mechanical device generally used for *raising liquids from lower level to a higher level.*
- It is achieved by creating a *low pressure at inlet & high pressure at outlet.*
- Low pressure cause liquid to lift from *sump to pump*, High pressure at outlet causes liquid to lift from *pump to required height*

The pumps are broadly classified into two as

- **Positive displacement Pumps** (Hydrostatic pumps)
- **Roto-dynamic pumps** (dynamic-pressure pumps)

Positive displacement pumps

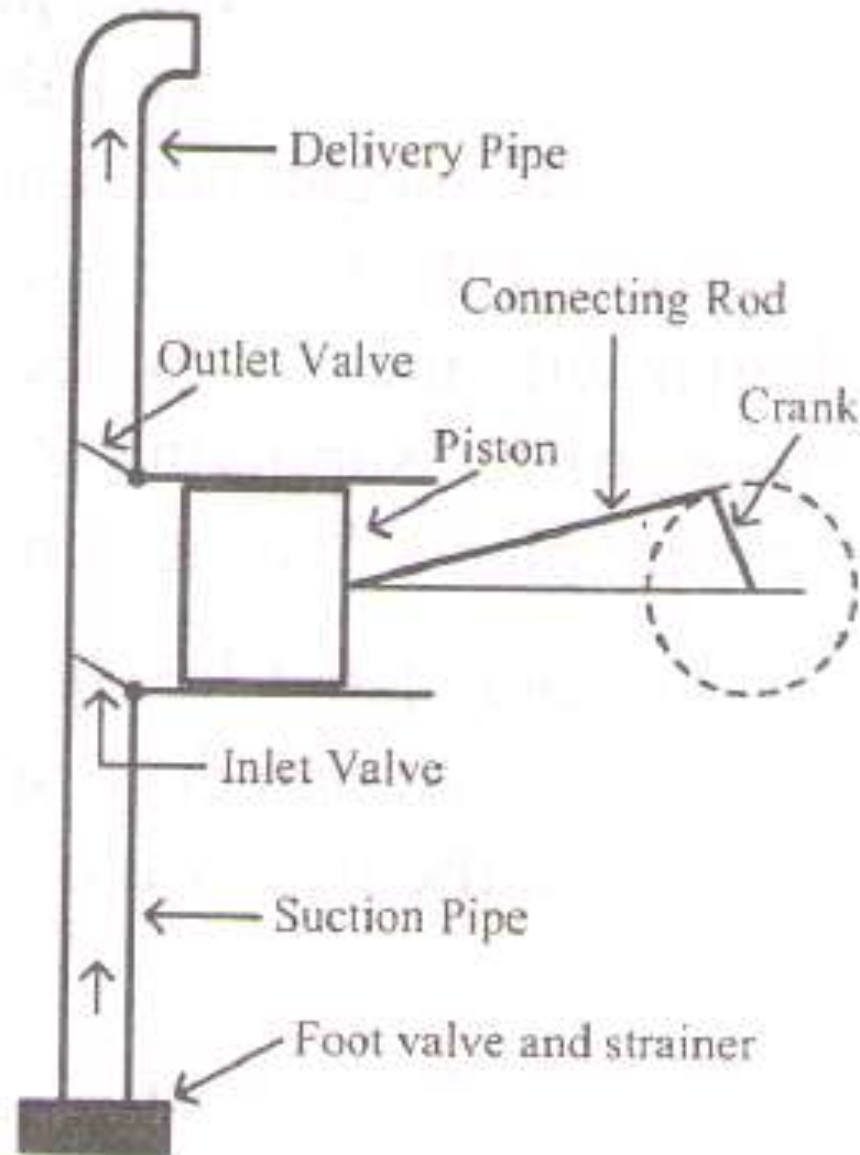
- Liquid is sucked and is actually pushed or displaced due to the thrust exerted on it by a ***moving (reciprocating) member*** which results in lifting the liquid to the required height
- These pumps work by alternately ***filling and emptying a cavity*** with the liquid to be pumped
- Discharge depends on speed of the pump
- Most common example is ***reciprocating pumps***

Rotodynamic pumps

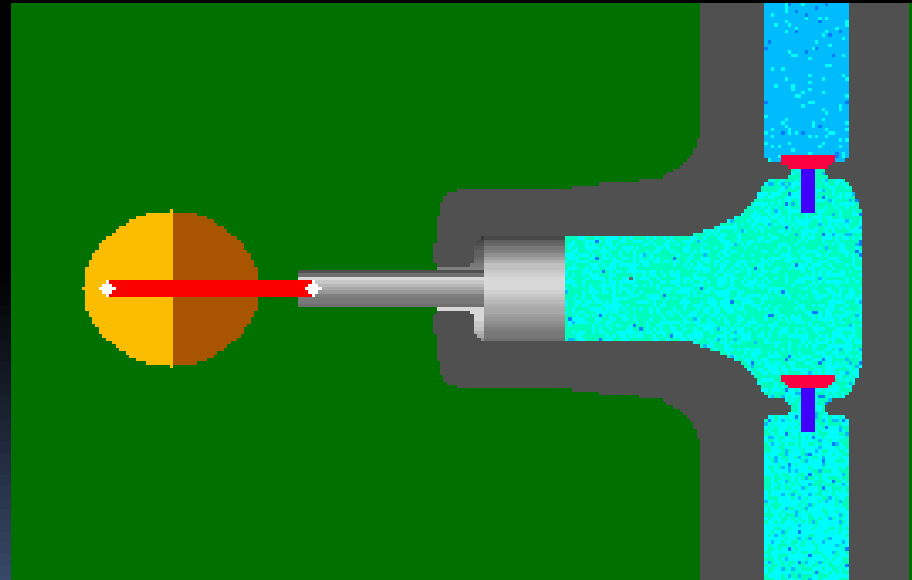
- These pumps are having a *rotating element* called *impeller* through which as the liquid passes, its angular momentum changes due to which the pressure energy of the liquid is increased
- A rotodynamic pump *does not push the liquid* as in a positive displacement pump
- *Centrifugal pump* is the most common example for a rotodynamic pump

Reciprocating Pumps

- Essentially consists of a *piston or plunger* which *moves to and fro* in a close fitting *cylinder*
- The cylinder is connected to *suction and delivery pipes*, each of which are provided with a non-return or one way valve called the *suction* and *delivery valve* respectively
- Suction and delivery valves open in *one direction only*
- The piston is connected to a *crank* by means of a *connecting rod*



- During the **outward stroke** of the piston a partial **vacuum is created** in the cylinder which enables the atmospheric pressure acting on the liquid surface to fill in the cylinder by forcing **open the suction valve**
- Due to the suction **water/fluid is drawn** into the cylinder through the suction valve.
- The **delivery valve** will be **closed** during this stroke.
- During the **return stroke**, the pressure of the fluid in the cylinder is increased which opens the delivery valve, closes the suction valve and pushes the fluid through the delivery valve.



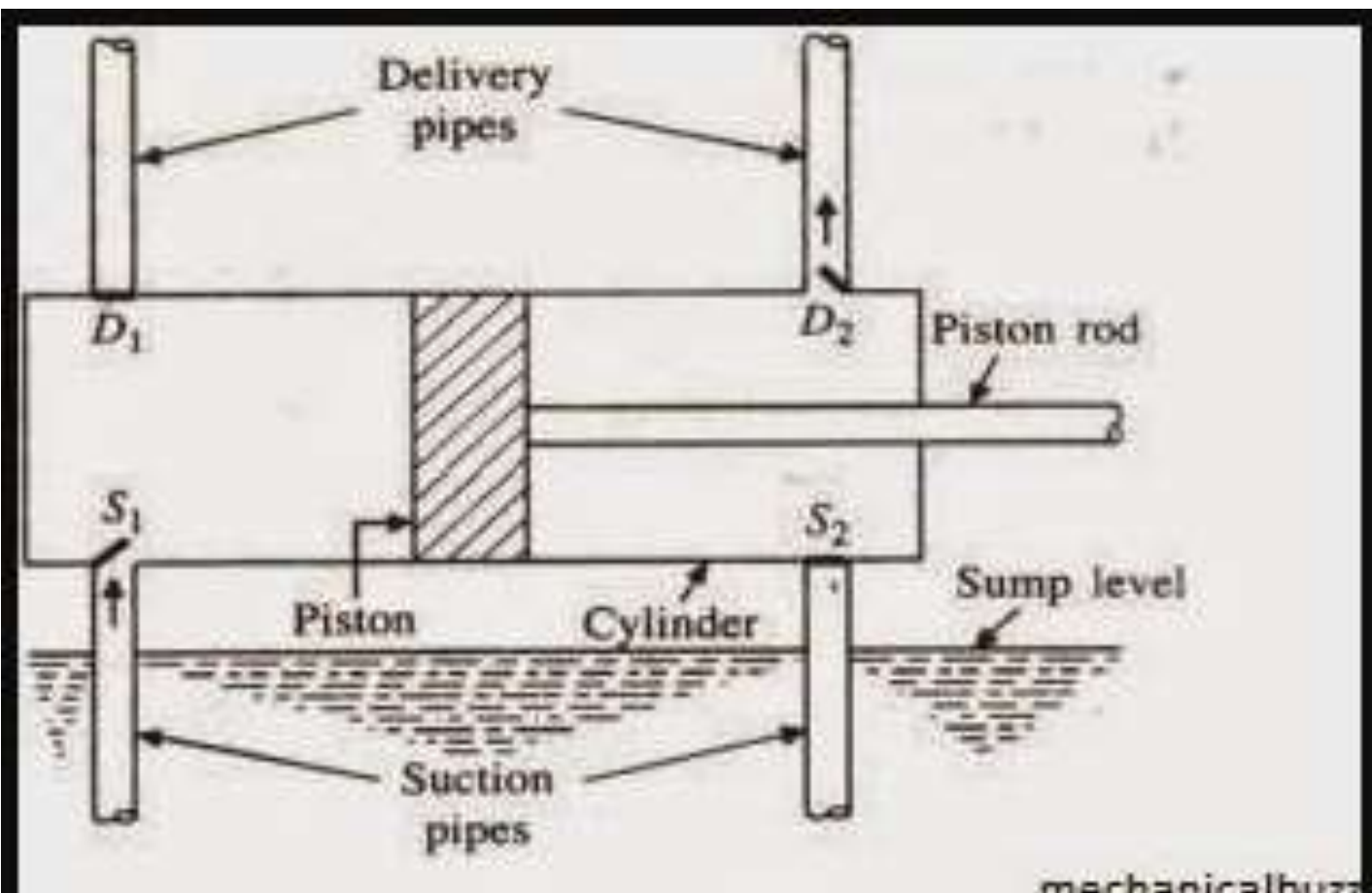
Types of reciprocating pumps

1. Single acting pump

- If the *liquid is in contact with one side of the piston*, it is a single acting pump
- It has only *one suction stroke and one delivery stroke* in one complete revolution of the crank

2. Double acting pump

- If the *liquid is in contact with both sides of the piston*, it is a double acting pump
- It has *two suction strokes and two delivery strokes* in one complete revolution of the crank

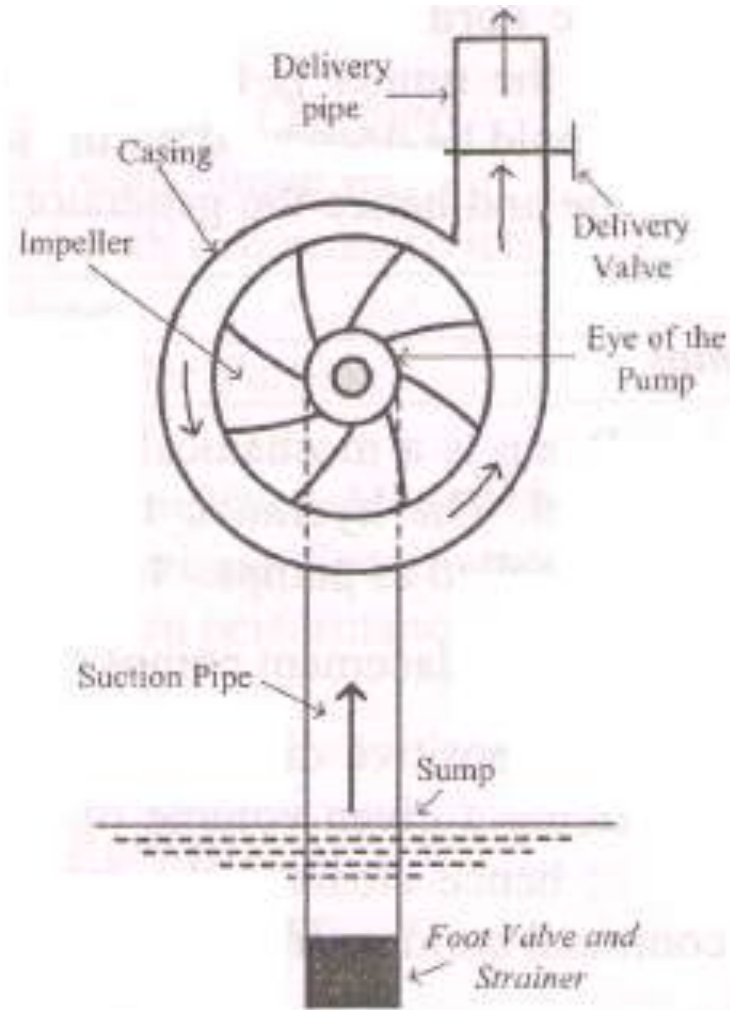


CENTRIFUGAL PUMP

- Centrifugal pumps are *rotodynamic type* of pumps.
- Low pressure at inlet & high pressure at outlet is obtained by centrifugal action of impeller
- The basic principle on which a centrifugal pump work is that *when a certain mass of liquid is made to rotate by an external force it is thrown away from the central axis* of rotation because a centrifugal head is impressed which enables the liquid to rise to higher level.

- Since the lifting of the liquid is due to the centrifugal action, these pumps are called *centrifugal pumps*
- In addition to the centrifugal action, the *angular momentum* of the liquid changes as it passes through the revolving wheel which also increases the pressure of liquid
- Gradually *increasing area of impeller* reduces velocity (Kinetic energy) of fluid which is converted in to useful *pressure energy*

- The main components of a centrifugal pump are
 - *impeller,*
 - *casing,*
 - *suction pipe with foot valve and strainer,*
 - *delivery pipe*
 - *delivery valve*
- The centrifugal pump is similar in construction with the Francis turbine
- But the difference is that the *fluid flow is in a direction opposite* to that in the turbine



CENTRIFUGAL PUMP



Parts of a centrifugal Pump

Impeller

- Impeller is a *circular wheel*, which is provided with a *series of curved vanes*, imparts energy into the fluid.
- The vanes are curved backward
- It is mounted on a shaft, which is coupled to an external source of energy, usually an electric motor, which imparts the required energy to the impeller there by making it to rotate.

Casing

- The impeller is surrounded by a *spiral (volute) shaped casing*. It is an airtight chamber.
- It is shaped in such a way that the liquid can flow through a passage of *gradually increasing area* with constant velocity.
- Partial *conversion of velocity energy into pressure energy* can take place in the casing.
- Moreover casing carries water from the impeller to the delivery pipe.
- In large centrifugal pumps air vent is provided on the casing. This is used to vent air at the time of priming

Suction pipe with foot valve and strainer

- The upper end of the suction pipe is connected to the casing at the center of the impeller.
- The lower end of suction pipe is fitted with a foot valve and strainer.
- The strainer keeps away the debris.
- The foot valve is a *non return* or one way type of valve which opens only in the upward direction

Delivery Pipe

- Delivery pipe is connected at its lower end to the outlet of the pump and it delivers the liquid to the required height.
- A delivery valve is fitted *near the outlet of the pump* to control the flow from the pump to the delivery pipe.

Working

- The first step in the working of centrifugal pump is **priming**.
- It is an operation by which suction pipe, casing of the pump and portion of delivery pipe up to the delivery valve is *filled with the liquid to be pumped, so as to remove air gaps*.
- The necessity of priming in centrifugal pump is due to the fact that the *centrifugal head generated by the impeller is directly proportional to the density of liquid* that is in contact with it.

- *After priming the delivery valve is kept closed* (to reduce starting torque of the motor) and the *electric motor starts to rotate with the impeller.*
- The *rotation of the impeller imparts a centrifugal head* to the liquid thereby *increasing pressure.*
- The pressure at any point is directly proportional to the square of the angular velocity (ω^2) and the distance of the point from the axis of rotation (r^2).
- Now the *delivery valve is opened and the liquid is allowed to flow in an outward radial direction.*
- *At the eye of the impeller a partial vacuum will be created*
- This causes the *liquid from the sump which is at atmospheric pressure to rush through the suction pipe* to the eye of the impeller thereby replacing the liquid which is being discharged from the entire circumference of the impeller.

Difference between Roto dynamic Pump and Positive Displacement Pump

Parameter	Positive displacement pump	Rotodynamic pump
Construction	More complex, consists of several moving parts.	Simple in construction.
Maintenance	High maintenance	Comparatively low
Suitability	Suitable for fairly low volumes of flow at high pressure	Suitable for large volumes of discharge at moderate pressure in a single stage
Fluid handling	Has higher efficiency while handling high viscous fluids.	Has lower efficiency while handling high viscous fluids.
Costs & Space considerations	More space and more initial expense. Less power consumption.	Less space and initial cost but more annual power consumption
Applications	Pump fluids even containing abrasive solids.	Usually used for pumping non-viscous fluids